

Students conducted an experiment to determine the changes in enthalpy ΔH° , entropy ΔS° , and Gibbs free energy ΔG° resulting from the dissociation of acetic acid (CH_3COOH) in water. The students first determined the acid dissociation constant K_a for the reaction by measuring the pH of an acetic acid solution under standard state conditions at various temperatures over a narrow range of less than 10 °C.

To perform the pH measurements, the acetic acid solution was initially cooled to 0 °C using an ice bath and then warmed with gentle heating to room temperature with continuous stirring. The pH was recorded at intervals of 1 °C. From the pH data, the value of K_a at each temperature T was calculated. The experimental results over the selected temperature range are shown in Table 1.

Table 1 Student Data for the pH and K_a of Acetic Acid with Increasing Temperature

T (°C)	T (K)	$1/T$ (K^{-1})	pH	K_a	$\ln K_a$
8	281	0.00356	2.63	5.495×10^{-6}	-12.11
9	282	0.00355	2.61	6.026×10^{-6}	-12.02
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To evaluate ΔH° and ΔS° , the students constructed a van 't Hoff plot (Figure 1), which is a graph of the natural log of the dissociation constant ($\ln K_a$) vs. the reciprocal absolute temperature ($1/T$).

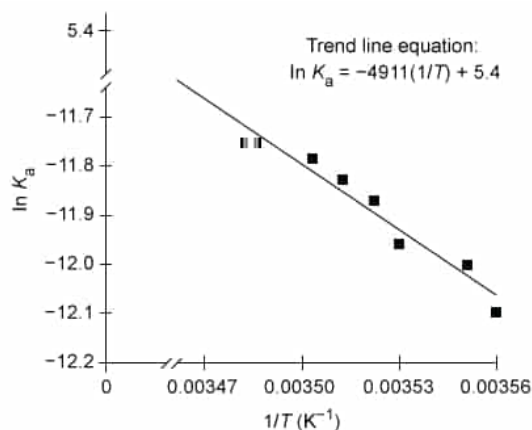


Figure 1 Linear van 't Hoff plot relating the dissociation constant K_a of acetic acid to temperature T

The relationship between K_a , T , ΔH° , and ΔS° in the van 't Hoff plot was then described using a linear fit of the data following the form of Equation 1, in which R is the gas constant ($8.3 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$).

$$\ln K_a = -\frac{\Delta H^\circ}{R} \left(\frac{1}{T} \right) + \frac{\Delta S^\circ}{R}$$

Equation 1

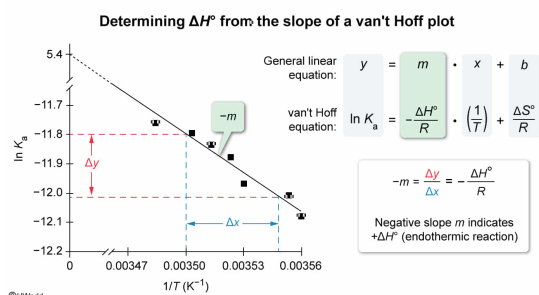
Based on the van 't Hoff plot shown in Figure 1, the dissociation of acetic acid is:

- ☐ A. endothermic, because the y-intercept of the trend line is positive, and therefore ΔS° is positive. [6%]
- ✓ ☒ B. endothermic, because the slope of the trend line is negative, and therefore ΔH° is positive. [80%]
- ☐ C. exothermic, because the slope of the trend line is positive, and therefore ΔH° is negative. [6%]
- ☐ D. exothermic, because the y-intercept of the trend line is negative, and therefore ΔS° is negative. [6%]

Correct

80% answered correctly

Explanation:



When measured under constant pressure, the **change in enthalpy** ΔH° for a reaction indicates the heat absorbed or released by the reaction. An **endothermic** reaction has a **positive ΔH°** because heat is *added* to the system (**heat absorbed** from the surroundings) whereas an **exothermic** reaction has a **negative ΔH°** because heat is produced by and *lost* from the system (**heat released** into the surroundings).

Exam questions sometimes test required topics by presenting concepts in an unfamiliar context using details that are not required knowledge but are explained within the passage. In this passage, a van 't Hoff plot enables the evaluation of ΔH° by presenting experimental data as a **linear function** in the form of

$$y = mx + b$$

where x and y are the x -axis and y -axis values, respectively, m is the slope of the line, and b is the y -intercept (ie, the point at which the line crosses the y -axis, when $x = 0$). When expressed in terms of the experimental data, the natural log of the **equilibrium constant**, $\ln K_a$, corresponds to y , the reciprocal absolute temperature $1/T$ corresponds to x , and m corresponds to $\Delta H^\circ/R$,

where R is the gas constant.

A van 't Hoff plot trendline in which the y -axis ($\ln K_a$) values decrease as the x -axis ($1/T$) values increase will have a negative slope. During the **dissociation** of acetic acid, $[\text{H}_3\text{O}^+]$ and $[\text{CH}_3\text{COO}^-]$ increase as the temperature increases. As a result, the equilibrium constant also

increases. Consequently, the van 't Hoff plot for the dissociation of acetic acid (Figure 1) has a negative slope ($-m$), which indicates that ΔH° is positive:

$$-m = -\frac{\Delta H^\circ}{R}$$

$$mR = +\Delta H^\circ$$

Therefore, the dissociation of acetic acid is endothermic.

(Choice A) Although the graph indicates an endothermic reaction, the reason is not due to the entropy, which is related to the y-intercept ($b = \Delta S^\circ/R$).

(Choices C and D) The trendline for an exothermic reaction would have a positive slope, indicating that ΔH° is negative. However, Figure 1 shows a trendline with a *negative* slope and a positive y-intercept, indicating that both ΔH° and ΔS° are positive.

Educational objective:

An endothermic reaction has a positive ΔH° and absorbs heat from the surroundings whereas an exothermic reaction has a negative ΔH° and releases heat into the surroundings.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402465

System:
Thermodynamics, Kinetics & Gas Laws

Students conducted an experiment to determine the changes in enthalpy ΔH° , entropy ΔS° , and Gibbs free energy ΔG° resulting from the dissociation of acetic acid (CH_3COOH) in water. The students first determined the acid dissociation constant K_a for the reaction by measuring the pH of an acetic acid solution under standard state conditions at various temperatures over a narrow range of less than 10 °C.

To perform the pH measurements, the acetic acid solution was initially cooled to 0 °C using an ice bath and then warmed with gentle heating to room temperature with continuous stirring. The pH was recorded at intervals of 1 °C. From the pH data, the value of K_a at each temperature T was calculated. The experimental results over the selected temperature range are shown in Table 1.

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To evaluate ΔH° and ΔS° , the students constructed a van 't Hoff plot (Figure 1), which is a graph of the natural log of the dissociation constant ($\ln K_a$) vs. the reciprocal absolute temperature ($1/T$).

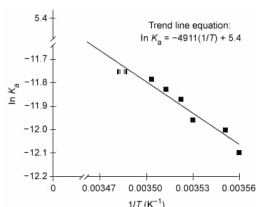


Figure 1 Linear van 't Hoff plot relating the dissociation constant K_a of acetic acid to temperature T

The relationship between K_a , T , ΔH° , and ΔS° in the van 't Hoff plot was then described using a linear fit of the data following the form of Equation 1, in which R is the gas constant ($8.3 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$).

$$\ln K_a = -\frac{\Delta H^\circ}{R} \left(\frac{1}{T} \right) + \frac{\Delta S^\circ}{R}$$

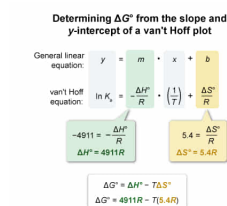
Equation 1

The slope of the trendline for the data shown in Figure 1 is -4911 and its y-intercept is 5.4. Which of the following is an expression for ΔG° at temperature T stated in terms of the data variables?

- ☒ A. $4911R - T(5.4R)$ [55%]
☐ B. $5.4R + T(4911R)$ [8%]
☐ C. $\frac{4911}{R} - T\left(\frac{5.4}{R}\right)$ [26%]
☐ D. $\frac{5.4}{R} + T\left(\frac{4911}{R}\right)$ [8%]

Correct
54% answered correctly

Explanation:



The change in **Gibbs free energy** ΔG° for a reaction is related to the changes in **enthalpy** ΔH° and **entropy** ΔS° at **temperature** T (measured in **Kelvin**) according to the formula:

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$$

In the equation above, the degree symbol indicates that all thermodynamic values are reported in **standard state** at equilibrium for a process in which the reactants have a **concentration of 1 M**. Note that temperature is not specified for standard state conditions in thermodynamics.

Although ΔH° describes the flow of heat in a reaction and ΔS° describes the disorder of the products relative to the reactants, ΔG° does not have a physical correlation. Rather, ΔG° is a **state function** that describes the **spontaneity** of the reaction (ie, its likelihood to occur without additional intervention).

A van 't Hoff plot enables the evaluation of ΔH° and ΔS° by presenting experimental data as a **linear function** in the form of

$$y = mx + b$$

where x and y are the x -axis and y -axis values, respectively, m is the slope of the line, and b is the y -intercept (ie, the point at which the line crosses the y -axis, when $x = 0$). When expressed in terms of Equation 1, m corresponds to $-\frac{\Delta H^\circ}{R}$ and b corresponds to $\frac{\Delta S^\circ}{R}$, where R is the gas constant. In this experiment, the slope of the trendline is -4911. Setting the m value of -4911 equal to $-\frac{\Delta H^\circ}{R}$ and solving for ΔH° gives:

$$m = -\frac{\Delta H^\circ}{R} = -4911$$

$$\Delta H^\circ = 4911R$$

Similarly, setting the y -intercept value of 5.4 equal to the y -intercept term $\frac{\Delta S^\circ}{R}$ and solving for ΔS° gives:

$$b = \frac{\Delta S^\circ}{R} = 5.4$$

$$\Delta S^\circ = 5.4R$$

Substituting the expressions for ΔH° and ΔS° above into the equation for Gibbs free energy yields:

$$\Delta G^\circ = 4911R - T(5.4R)$$

(Choice B) This answer is obtained when ΔS° and ΔH° are switched in the Gibbs free energy equation.

(Choice C) This answer is obtained if the terms for the slope and the y -intercept ($-4911/R$ and $5.4/R$) from Equation 1 are substituted into the Gibbs free energy equation without first solving the respective terms for ΔH° and ΔS° .

(Choice D) This answer is obtained by mixing up the substitution of the terms for the slope and the y -intercept from Equation 1 into the Gibbs free energy equation and failing to first solve the respective terms for ΔH° and ΔS° .

Educational objective:

The Gibbs free energy is a state function that describes the spontaneity of a reaction. It is related to the temperature, enthalpy, and entropy of the reaction via the equation $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$, where T is given in Kelvin.

Time spent: 0 sec
Subject:
General Chemistry

QID: 402466
System:
Thermodynamics, Kinetics & Gas Laws

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To perform the pH measurements, the acetic acid solution was initially cooled to 0°C using an ice bath and then warmed with gentle heating to room temperature with continuous stirring. The pH was recorded at intervals of 1°C . From the pH data, the value of K_a at each temperature T was calculated. The experimental results over the selected temperature range are shown in Table 1.

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To evaluate ΔH° and ΔS° , the students constructed a van 't Hoff plot (Figure 1), which is a graph of the natural log of the dissociation constant ($\ln K_a$) vs. the reciprocal absolute temperature ($1/T$).

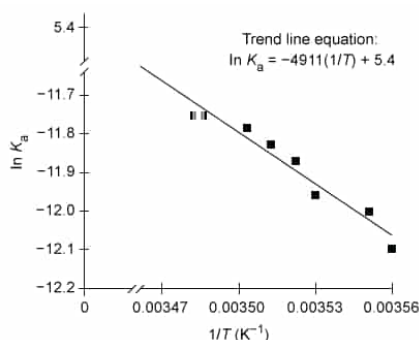


Figure 1 Linear van 't Hoff plot relating the dissociation constant K_a of acetic acid to temperature T

The relationship between K_a , T , ΔH° , and ΔS° in the van 't Hoff plot was then described using a linear fit of the data following the form of Equation 1, in which R is the gas constant ($8.3 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$).

$$\ln K_a = -\frac{\Delta H^\circ}{R}\left(\frac{1}{T}\right) + \frac{\Delta S^\circ}{R}$$

Equation 1

Based on the experimental results in the passage, the dissociation of acetic acid at temperatures outside of the 281 K to 287 K range will be:

- I. spontaneous at all possible temperatures.
- II. only nonspontaneous at lower temperatures.
- III. only spontaneous at higher temperatures.

- ☐ A. I only [17%]
- ☐ B. II only [10%]
- ☐ C. III only [36%]
- ☒ D. II and III only [35%]

Correct

36% answered correctly

Explanation:

Effect of temperature on Gibbs free energy				
$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$				
$\Delta G^\circ < 0$ spontaneous (thermodynamically favorable) $\Delta G^\circ > 0$ nonspontaneous (thermodynamically unfavorable)				
ΔH°	ΔS°	$-T\Delta S^\circ$	ΔG° with rising temperature	Spontaneity with rising temperature (T)
-	+	-	Increasingly negative	Always spontaneous
+	-	+	Increasingly positive	Always nonspontaneous
-	-	+	Decreasingly negative then increasingly positive	Spontaneous at lower T, nonspontaneous at higher T
+	+	-	Decreasingly positive then increasingly negative	Spontaneous at higher T, nonspontaneous at lower T

The spontaneity (or nonspontaneity) of a process is evaluated using the change in **standard Gibbs free energy ΔG°** , which is related to the changes in **enthalpy ΔH°** , **entropy ΔS°** , and the absolute temperature T of the system according to the equation:

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$$

Accordingly, when $\Delta G^\circ < 0$, the reaction is **spontaneous** (ie, energetically favorable) because the products have less free energy than the reactants. When $\Delta G^\circ > 0$, the reaction is **nonspontaneous** (ie, energetically unfavorable) because the products have more free energy than the reactants. When $\Delta G^\circ = 0$, the reaction is already at its lowest energy state at **equilibrium**.

As such, four possible outcomes can occur:

- When ΔS° is positive and ΔH° is negative (ie, an **exothermic** reaction), the process is always spontaneous ($-\Delta G^\circ$).
- When ΔS° is negative and ΔH° is positive (ie, an **endothermic** reaction), the process is nonspontaneous ($+\Delta G^\circ$) at any temperature.
- When both ΔS° and ΔH° are negative, the process is spontaneous ($-\Delta G^\circ$) at lower temperatures and nonspontaneous ($+\Delta G^\circ$) at higher temperatures.
- When both ΔS° and ΔH° are positive, the process is spontaneous ($-\Delta G^\circ$) at higher temperatures and nonspontaneous ($+\Delta G^\circ$) at lower temperatures.

Based on the van 't Hoff plot provided in Figure 1 of the passage, ΔH° and ΔS° are **both positive**; therefore, the reaction will be nonspontaneous at lower temperatures (**Number II**) but spontaneous at higher temperatures (**Number III**).

(Number I) The reaction is *not* spontaneous at all possible temperatures because it is endothermic at low temperatures.

Educational objective:

A spontaneous process does not require a continuous input of additional energy. The change in Gibbs free energy (ΔG°) is evaluated as $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ and is used to identify whether a process is spontaneous ($\Delta G^\circ < 0$), nonspontaneous ($\Delta G^\circ > 0$), or at equilibrium ($\Delta G^\circ = 0$). Processes in which both changes in entropy ΔS° and enthalpy ΔH° are positive are spontaneous at higher temperatures and nonspontaneous at lower temperatures.

Time spent: 0 sec
Subject:
General Chemistry

QID: 402467
System:
Thermodynamics, Kinetics & Gas Laws

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To perform the pH measurements, the acetic acid solution was initially cooled to 0 °C using an ice bath and then warmed with gentle heating to room temperature with continuous stirring. The pH was recorded at intervals of 1 °C. From the pH data, the value of K_a at each temperature T was calculated. The experimental results over the selected temperature range are shown in Table 1.

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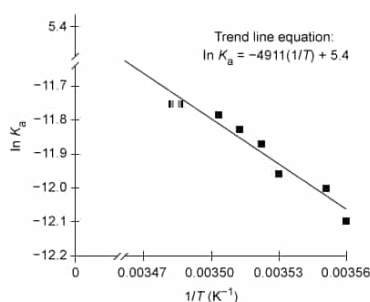


Figure 1 Linear van 't Hoff plot relating the dissociation constant K_a of acetic acid to temperature T

The relationship between K_a , T , ΔH° , and ΔS° in the van 't Hoff plot was then described using a linear fit of the data following the form of Equation 1, in which R is the gas constant ($8.3 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$).

$$\ln K_a = -\frac{\Delta H^\circ}{R}\left(\frac{1}{T}\right) + \frac{\Delta S^\circ}{R}$$

Equation 1

Based on the data presented in Table 1, as the pH of the acetic acid solution decreases when going from a temperature of 281 K to 287 K, ΔG° for the dissociation of acetic acid:

- ✔ ☒ A. decreases. [55%]
☐ B. increases. [38%]
☐ C. is constant. [5%]
☐ D. is zero.

Correct

55% answered correctly

Explanation:

The change in Gibbs free energy ΔG° for the dissociation of acetic acid

T (°C)	T (K)	$1/T$ (K ⁻¹)	pH	K_a	$\ln K_a$
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ΔG° decreases with decreasing pH (increasing T and K_a)

$\Delta G^\circ = -RT \ln K_a$

The change in **standard Gibbs free energy ΔG°** of a reaction is related to its **equilibrium constant K_{eq}** at a given **absolute (Kelvin)**

temperature T by the gas constant R according to the equation:

$$\Delta G^\circ = -RT \ln K_{\text{eq}}$$

In the case of the dissociation of a weak acid, the equilibrium constant of the reaction is called the acid dissociation constant, K_a .

According to the data provided in Table 1, as the temperature increases, the pH of the acetic acid solution decreases, and K_a for the dissociation of acetic acid increases. This corresponds to a decrease in ΔG° . For example, when the pH of the solution is 2.63, ΔG° is equal to:

$$\Delta G^\circ = -R(281 \text{ K})(-12.11) = R(3402 \text{ K})$$

And when the pH of the solution is 2.55, ΔG° is equal to

$$\Delta G^\circ = -R(287 \text{ K})(-11.74) = R(3369 \text{ K})$$

Therefore, ΔG° **decreases** as the pH of the solution decreases with increasing temperature.

(Choice B) In Table 1, the pH decreases and K_a increases as the temperature increases. Because $\Delta G^\circ = -RT \ln K_a$, ΔG° *decreases* with decreasing pH and increasing temperature.

(Choice C) Given that $\Delta G^\circ = -RT \ln K_a$, ΔG° cannot stay the same because both the temperature and the equilibrium constant are changing in this experiment.

(Choice D) Based on the data provided in Table 1, ΔG° cannot equal zero over the experimental temperature range.

Educational objective:

The change in Gibbs free energy ΔG° for a reaction at a given temperature is related to the equilibrium constant K_{eq} by the relationship $\Delta G^\circ = -RT \ln K_{\text{eq}}$.

Time spent: 0 sec
Subject:
General Chemistry

QID: 402468
System:
Thermodynamics, Kinetics & Gas Laws

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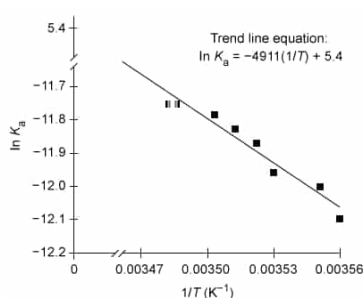


Figure 1 Linear van 't Hoff plot relating the dissociation constant K_a of acetic acid to temperature T

The relationship between K_a , T , ΔH° , and ΔS° in the van 't Hoff plot was then described using a linear fit of the data following the form of Equation 1, in which R is the gas constant ($8.3 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$).

$$\ln K_a = -\frac{\Delta H^\circ}{R} \left(\frac{1}{T} \right) + \frac{\Delta S^\circ}{R}$$

Equation 1

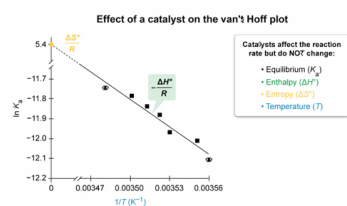
Would changes in the van 't Hoff plot be observed if the reaction rate were increased by adding a catalyst during the experiment?

- ☐ A. No, because the initial concentration of acetic acid does not change. [3%]
- ☒ B. No, because ΔG° , ΔH° , and ΔS° are independent of the reaction rate. [83%]
- ☐ C. Yes, because the reaction rate and ΔG° depend on temperature. [7%]
- ☐ D. Yes, because spontaneity depends on the reaction rate. [4%]

Correct

84% answered correctly

Explanation:



Thermodynamic values indicate if a reaction is **spontaneous** or not, but they **do not** indicate the **rate** of a reaction. Enthalpy, entropy, and Gibbs free energy are **state functions**, which mean they describe the energetic differences between the reactants and the products in their current state. These values are **independent of the chemical pathway** that a reaction takes to get from reactants to products.

A **catalyst** is a chemical species that is added to a reaction to increase the rate of the reaction by stabilizing the **transition state** and decreasing the activation energy of the reaction. A catalyst does not change the amount of products produced and does not change the relative energies of the reactants and products.

The van 't Hoff plot for the experiment described in the passage is used to find the thermodynamic values for the dissociation of acetic acid.

The results are based on the final state of the reactants and products at a given temperature, not the reaction mechanism pathway.

Increasing the rate of the reaction by adding a catalyst will not affect the amount of product (H^+) that is produced. Therefore, a catalyst will not change the pH of the solution or the thermodynamic parameters measured in the experiment.

(Choice A) Although the addition of a catalyst will not change the initial concentration of acetic acid, the thermodynamic results in Figure 1 are from state functions that depend on the reaction conditions rather than on the initial amount of acetic acid or the specific reaction pathway invoked by a catalyst.

(Choice C) Although it is true that the reaction rate and ΔG° are both affected by temperature, changing the reaction rate will not change the van 't Hoff plot because ΔS° and ΔH° are independent of temperature.

(Choice D) Spontaneity indicates whether a reaction will proceed without the continuous input of additional energy. Spontaneity does not indicate how long the process will take.

Educational objective:

Thermodynamic values indicate if a reaction is spontaneous or not, but they do not indicate the rate of a reaction because thermodynamic values are state functions that are independent of the reaction pathway.

Time spent:0 sec

Subject:
General Chemistry

QID:402469

System:
Thermodynamics, Kinetics & Gas Laws

Students conducted an experiment to determine the changes in enthalpy ΔH° , entropy ΔS° , and Gibbs free energy ΔG° resulting from the dissociation of acetic acid (CH_3COOH) in water. The students first determined the acid dissociation constant K_a for the reaction by measuring the pH of an acetic acid solution under standard state conditions at various temperatures over a narrow range of less than 10 °C.

To perform the pH measurements, the acetic acid solution was initially cooled to 0 °C using an ice bath and then warmed with gentle heating to room temperature with continuous stirring. The pH was recorded at intervals of 1 °C. From the pH data, the value of K_a at each temperature T was calculated. The experimental results over the selected temperature range are shown in Table 1.

Table 1 Student Data for the pH and K_a of Acetic Acid with Increasing Temperature

T (°C)	T (K)	$1/T$ (K^{-1})	pH	K_a	$\ln K_a$
8	281	0.00356	2.63	5.495×10^{-6}	-12.11
9	282	0.00355	2.61	6.026×10^{-6}	-12.02
10	283	0.00353	2.60	6.310×10^{-6}	-11.97
11	284	0.00352	2.58	6.918×10^{-6}	-11.88
12	285	0.00351	2.57	7.244×10^{-6}	-11.84
13	286	0.00350	2.56	7.586×10^{-6}	-11.79
14	287	0.00348	2.55	7.942×10^{-6}	-11.74

To evaluate ΔH° and ΔS° , the students constructed a van 't Hoff plot (Figure 1), which is a graph of the natural log of the dissociation constant ($\ln K_a$) vs. the reciprocal absolute temperature ($1/T$).

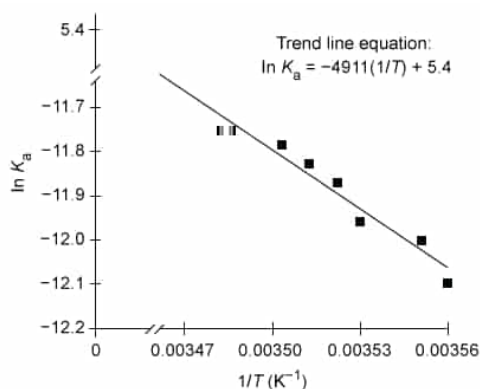


Figure 1 Linear van 't Hoff plot relating the dissociation constant K_a of acetic acid to temperature T

The relationship between K_a , T , ΔH° , and ΔS° in the van 't Hoff plot was then described using a linear fit of the data following the form of Equation 1, in which R is the gas constant ($8.3 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$).

$$\ln K_a = -\frac{\Delta H^\circ}{R} \left(\frac{1}{T} \right) + \frac{\Delta S^\circ}{R}$$

Equation 1

During the dissociation of acetic acid at the temperatures tested in Table 1, the molecular disorder within the reaction vessel:

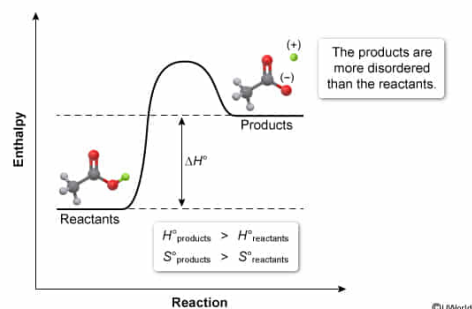
- ☐ A. increases, and the enthalpy of the reactants is greater than the enthalpy of the products. [27%]
- ✓ ☒ B. increases, and the enthalpy of the reactants is less than the enthalpy of the products. [57%]
- ☐ C. decreases, and the enthalpy of the reactants is greater than the enthalpy of the products. [9%]
- ☐ D. decreases, and the enthalpy of the reactants is less than the enthalpy of the products. [5%]

Correct

57% answered correctly

Explanation:

Reaction energy diagram for the dissociation of acetic acid



In an **endothermic reaction**, heat is absorbed by the system from the surroundings as the reaction proceeds. As a result, an endothermic reaction has a **positive change in enthalpy** ΔH° , which indicates that the **products** have **higher enthalpy** than the **reactants** because ΔH° is evaluated as:

$$\Delta H^\circ = H^\circ_{\text{products}} - H^\circ_{\text{reactants}}$$

However, knowing the relative enthalpy of the reactants and products does not directly give information about the **change in entropy** ΔS° of the reaction. An endothermic reaction can result in either an increase or a decrease in entropy depending on the change in the extent of disorder within the system.

Dissociation reactions are typically endothermic processes because energy must be added to the system (as heat) to dissociate (break) the bonds in the reactants. Furthermore, dissociation reactions tend to increase entropy because a larger molecule is broken into two (or more) smaller molecules, which generally increases the disorder of the system.

Based on the results from the trendline in Figure 1, the dissociation of acetic acid yields a **positive** ΔH° and ΔS° . A positive ΔH° indicates that $H^\circ_{\text{products}}$ is **greater** than $H^\circ_{\text{reactants}}$, and a positive ΔS° indicates that the molecular disorder of the system **increases**.

(Choice A) Although entropy does increase, the enthalpy of the products is *not* lower than the enthalpy of the reactants. This would result in a **negative** ΔH° , which would occur only for an **exothermic reaction**.

(Choices C and D) Entropy *increases* in this reaction because the dissociation reaction breaks up a larger molecule into two smaller, charged molecules. The increasing entropy is verified graphically on the van 't Hoff plot as a positive y-intercept.

Educational objective:

Endothermic reactions have a positive ΔH° and absorb heat from the surroundings as the reaction proceeds because the products have higher energy than the reactants. An endothermic reaction can result in either an increase or a decrease in entropy, depending on the change in the extent of disorder within the system.

Time spent: 0 sec

Subject:
General Chemistry

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System:
Thermodynamics, Kinetics & Gas Laws